UTI Microfluidic Cartridge Diagnostic Report

Scan ID: scan_healthy_03 | Date: 21/9/2026, 12:48:18 am | Batch: GENEX-B26-HEALTHY

Patient: Clara Bennet (35y / Female)
Facility: Preventive Care Medical Group - Wellness Pavilion
Chief Complaint: Routine annual physical exam. Completely asymptomatic.

MRN: MRN-1029384 | Specimen: SP-2026-1102F
Clinician: Dr. James Harrison, MD
Collection: Routine Clean-Catch Urine

DIAGNOSTIC STATUS: NEGATIVE / HEALTHY BASELINE — NO ACTIVE UTI

Normal leukocyte esterase activity (<10 WBC/µL) and absence of detectable uropathogens across all aptamer panel channels. Broad susceptibility profile observed. No antimicrobial therapy indicated.

1. Pyuria / Pus-Cell Assay (Layer 1 — Leukocyte Esterase)

INFECTION SEVERITY	PYURIA WBC ESTIMATE	SIGNAL OPTICAL DENSITY
NEGATIVE	< 10 WBC/µL (Normal Physiological Range)	0.012 OD

2. 4-Plex Pathogen Aptamer Panel (Layer 2 — Aptamer Capture)

TARGET ORGANISM	GRAM STAIN	OPTICAL DENSITY	STATUS
Escherichia coli	Gram-negative	0.020 (Cutoff: 0.22)	Not Detected
Klebsiella pneumoniae	Gram-negative	0.018 (Cutoff: 0.2)	Not Detected
Proteus mirabilis	Gram-negative	0.015 (Cutoff: 0.2)	Not Detected
Pseudomonas aeruginosa	Gram-negative	0.016 (Cutoff: 0.21)	Not Detected

3. Antibiotic Susceptibility Testing (Layer 3 — 4× Replicate Panel)

ANTIMICROBIAL AGENT	CLASS / LINE	MEAN OD ± SD	CONCORDANCE	RESULT
Ampicillin	Aminopenicillin	0.045 ± 0.01	4/4 concordant (100%)	**SENSITIVE**
Ciprofloxacin	Fluoroquinolone	0.040 ± 0.01	4/4 concordant (100%)	**SENSITIVE**
Nitrofurantoin* (1st-Line)	Nitrofuran	0.038 ± 0.01	4/4 concordant (100%)	**SENSITIVE**
Cotrimoxazole* (1st-Line)	Folate Synthesis Inhibitor	0.042 ± 0.01	4/4 concordant (100%)	**SENSITIVE**
Fosfomycin* (1st-Line)	Phosphonic Acid Derivative	0.035 ± 0.01	4/4 concordant (100%)	**SENSITIVE**

ANTIMICROBIAL STEWARDSHIP & FIRST-LINE RECOMMENDATIONS:
No active uropathogen or significant pyuria detected. Antimicrobial therapy is not indicated under clinical stewardship guidelines. Hydration and observation recommended.

LEGAL MEDICAL DISCLAIMER: Genex is an investigational point-of-care software reader for microfluidic cartridges. This report does not constitute a certified medical diagnosis and is intended to support clinical decision-making. Treatment must be confirmed by a licensed medical practitioner in accordance with local antimicrobial guidelines.